Cosmology: Cosmology Lecture 8

Leonard Susskind

Transcript-derived notes curated by LazyingArt LLC with Video2Book

Original lecture by Leonard Susskind. Transcript-derived notes curated by LazyingArt LLC with Video2Book.

Chapter 1

Baryogenesis and the Prelude to Inflation

We were promised inflation, but the lecture first closes the baryogenesis story, and that order is exactly right. Susskind even jokes that baryogenesis can be finished quickly because so little is known in detail. The joke is instructive. What we know here is not a finished theory but a set of structural requirements. The evening is organized around two small numbers: first the baryon excess, of order 10^{-8}, and then the primordial lumpiness, of order 10^{-5}. These notes follow Leonard Susskind's progression closely, preserving the order in which the questions are sharpened and the mathematics is introduced; where a blackboard equation or sketch matters, we keep supporting lecture frames drawn from the LazyingArt LLC and Video2Book curation.

1.1 From Entropy to the Baryon Excess

The lecture begins with a historical question due to Bob Wagoner: why is the entropy of the universe so large? In the context of blackbody radiation, entropy is, up to factors of order one, essentially counted by the number of photons. So one can phrase the question as: why are there so many photons for every proton and electron?

Susskind's point is that this question is upside down. In the very early universe, when the plasma was hot and in thermal equilibrium, protons and antiprotons were both present in enormous numbers, and each was of the same order as the number of photons. The same was true of electrons and positrons. The photons did not have to be manufactured out of a tiny residue of matter. Matter and antimatter were both abundant, and almost all of them later annihilated.

That inversion is the first real move of the lecture. The problem is not the abundance of photons. It is the tiny mismatch that survived the annihilation era:

$$\frac{N_p - N_{\bar{p}}}{N_\gamma} \sim 10^{-8}, \qquad N_\gamma \sim N_p + N_{\bar{p}} \quad \text{in the early thermal plasma.} \tag{1.1}$$

So the excess of baryons over antibaryons was only about one part in 10^8 of the original baryon-antibaryon population. Once the question is expressed that way, it ceases to be a vague cosmological fact and becomes a quantitative puzzle. Susskind emphasizes exactly this point: once there is an oddball number on the board, one wants to know why that number is what it is.

The lecture also gives a short historical remark that matters for tone. Susskind and Savas Dimopoulos independently rediscovered the relevant conditions, but the priority belongs to Sakharov's 1967 paper. So the modern name is the right one: these are the Sakharov conditions. The chapter should preserve that modesty. We are not deriving a finished baryogenesis model. We are isolating the conditions under which any such model could work.

1.2 Baryon Number, Conservation, and the Proton-Decay Thought Experiment

Susskind now begins with a hypothesis that he plainly regards as useful but probably wrong: suppose baryon number were exactly conserved. Before testing that idea, he pauses to define the quantity.

Baryon number may be described in quark language. The lecture corrects itself in real time here: the net number of quarks minus antiquarks is not the baryon number itself, but three times it. The cleaned definition is therefore

$$B = \frac{N_q - N_{\bar{q}}}{3}. \tag{1.2}$$

This normalization ensures that a proton, which contains three quarks, carries baryon number 1.

It is important that baryon number is only *like* electric charge in one respect and unlike it in another. If it is conserved, then it is frozen in the same bookkeeping sense that electric charge is frozen in: it does not change with time. But it is not the source of a long-range Coulomb field. Protons make electric fields because they are electrically charged, not because they carry baryon number.

If baryon number were exactly conserved, any initial excess would simply remain. That is the content of the hypothesis: the baryon excess would not be created dynamically; it would merely be an initial condition carried forward unchanged. The lecture then asks what must fail if we want a dynamical explanation.

The cleanest imagined violation is proton decay. A proton is stable for all practical purposes in the present universe, but if it were to decay into lighter stable objects while conserving electric charge, the most natural schematic channel is

$$p \to e^+ + \gamma. \tag{1.3}$$

This is the first concrete image of baryon-number violation.

The blackboard frame is only the beginning of the sketch, so the cleaned note version should display the channels that the lecture actually uses:

A student asks whether a neutrino could appear in the final state, and the lecture briefly notes that one must also keep track of fermion parity: a proton is a fermion, so the final state must contain an odd number of fermions. But the main point is not spectroscopy. The main point is that a simple process can be imagined in which baryon number changes.

At first approximation this process is not something we see. Protons are extraordinarily stable. That is precisely why the lecture chooses it: not because it is an observed decay channel, but because it is the cleanest thought experiment for what baryon-number violation would look like. From here the first Sakharov condition is immediate: if the universe began with no matter-antimatter bias, then baryon number must be violable in order for any excess to be produced dynamically at all.

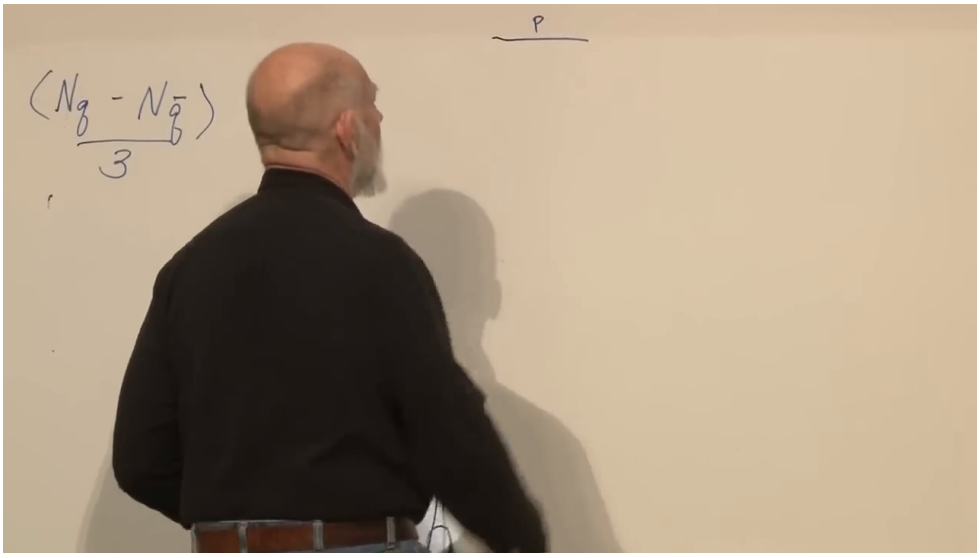

Figure 1.1: Opening line of the blackboard proton-decay sketch. The frame records the board sequence, while the completed decay channel is reconstructed from the transcript rather than from the incomplete drawing alone.

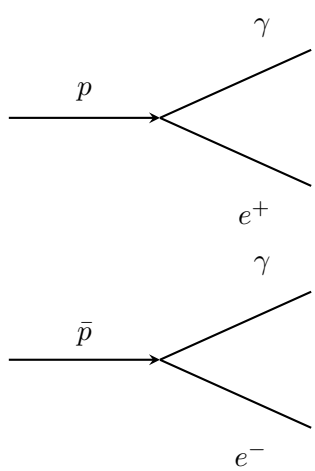

Figure 1.2: Cleaned schematic channels used in the baryogenesis argument. The upper process lowers baryon number; the lower charge-conjugate process raises it back.

1.3 Why Baryon Violation Alone Is Not Enough

Before going on, the lecture pauses over the obvious empirical obstacle. If a proton lives longer than 10^{32} or 10^{33} years, why should proton decay matter at all in cosmology? The answer is environmental. In unified pictures one imagines heavy virtual particles deep inside the Feynman diagram; schematically the amplitude is suppressed by inverse powers of a large mass M. For a cold proton at rest that suppression makes the process fantastically rare. But the early universe was hot. Quarks and baryons were constantly kicked around by very high-energy collisions, and in that environment the suppression can be much less severe. So the first Sakharov condition need not contradict the empirical longevity of present-day protons.

Now the lecture tightens the argument. The first Sakharov condition is necessary, but it is not sufficient. Suppose processes like $p \to e^+ + \gamma$ are allowed. Then the charge-conjugate process

$$\bar{p} \to e^- + \gamma \tag{1.4}$$

is also allowed. If the laws of physics treat particles and antiparticles symmetrically, then on average there will be as many decays of one sort as of the other. In that case baryon number changes, but it does not change with a preferred sign.

The lecture also insists on another point. In the very early universe these processes do not run only one way. The plasma contains electrons, positrons, and photons in abundance, so inverse reactions also occur. That is why one must not mistake baryon-number violation for a one-way machine that automatically manufactures matter. By itself it only gives statistical traffic in and out of the baryon sector.

This is the first place where the lecture naturally stops and answers the obvious objection.

1.3.1 Question & Answer

Why can the observed baryon excess not be just a statistical fluctuation?

Because the numbers are completely wrong for that explanation. If a variable is produced by random statistics, its typical fluctuation is of order the square root of the number of trials:

$$\Delta N \sim \sqrt{N}, \qquad \frac{\Delta N}{N} \sim \frac{1}{\sqrt{N}}. \tag{1.5}$$

For the observable universe, the lecture uses the rough estimate

$$N \sim 10^{80}. \tag{1.6}$$

Therefore a purely statistical baryon excess would be of relative size

$$\frac{\Delta N}{N} \sim 10^{-40}, \tag{1.7}$$

whereas the observed excess is of order 10^{-8}. The actual asymmetry is larger by thirty-two orders of magnitude.

There is also a spatial problem. A statistical fluctuation would not naturally give the same sign everywhere. One would expect matter-dominated patches here and antimatter-dominated patches there. The lecture points to the absence of antinuclei in cosmic rays, and to the absence of the

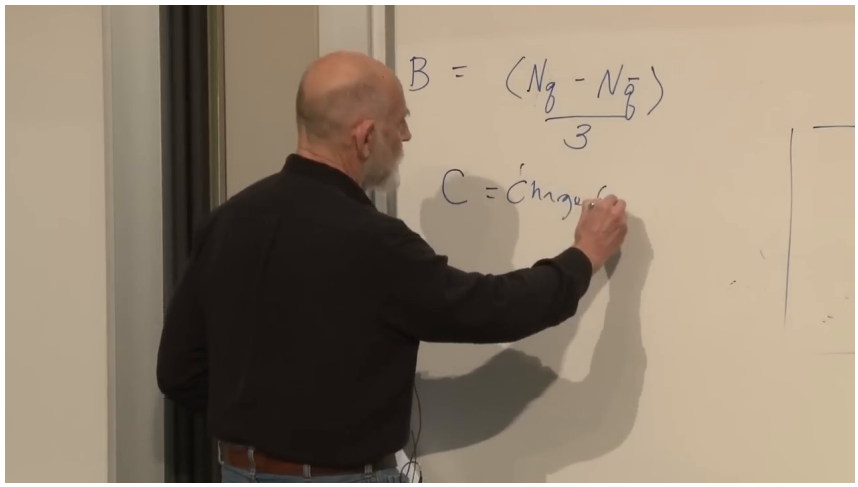

Figure 1.3: Baryon number on the upper line, followed by the introduction of C as charge conjugation symmetry. The frame is useful precisely because it preserves the transition from baryon counting to symmetry language.

enormous annihilation signals that neighboring matter and antimatter galaxies would produce, as evidence that the universe is not divided into such alternating domains.

So baryon-number violation alone gives, at best, a tiny fluctuation. It does not give the observed, coherent excess. We now know what the next ingredient must be: a bias in the laws themselves.

1.4 C, P, T, and the Need for a Bias

At this point the lecture pivots to the language of discrete symmetries. The order matters. First comes charge conjugation C, then parity P, then time reversal T, and only after that the more serious theorem involving their product.

The blackboard transition is worth preserving explicitly:

$$B = \frac{N_q - N_{\bar{q}}}{3}, \qquad C = \text{charge conjugation symmetry.} \tag{1.8}$$

Charge conjugation exchanges particles with antiparticles. Parity is mirror reflection, or left-right interchange. Time reversal, at the most naive level, asks whether a movie of a microscopic process run backward is still a possible microscopic process. In compact notation,

$$C: \ \text{particle} \leftrightarrow \text{antiparticle}, \qquad P: \ \text{left} \leftrightarrow \text{right}, \qquad T: \ t \mapsto -t. \tag{1.9}$$

The lecture then makes the standard modern point: neither C nor P nor T separately is the deep exact symmetry. The exact theorem of relativistic quantum field theory is CPT.

There is a short Schrödinger-equation digression here, and it belongs exactly here. It is not a side remark. It explains why time reversal is slightly subtler than simply replacing t by $-t$. The blackboard only shows the beginning of the equation, so we keep the cue schematic and accompany it by the standard cleaned transformation:

$$i\,\frac{\partial \psi}{\partial t} = \cdots, \qquad T: \ \psi(t) \mapsto \psi^*(-t). \tag{1.10}$$

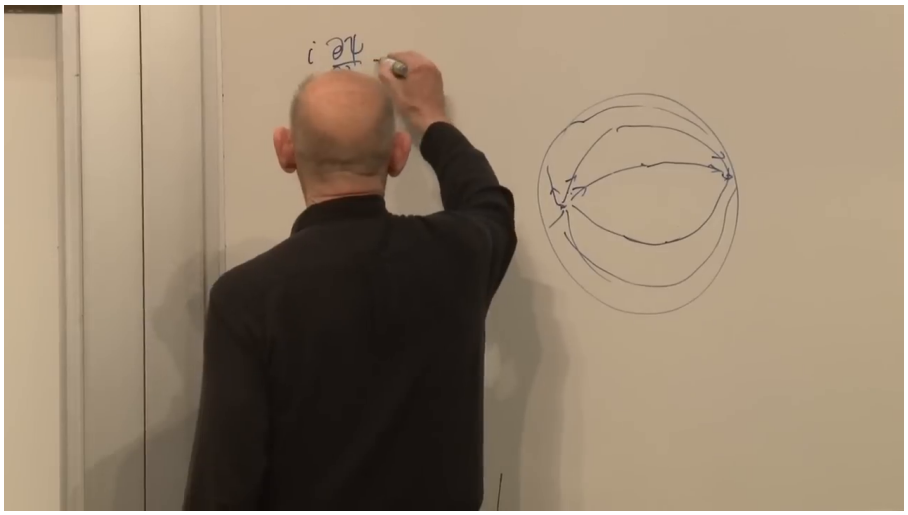

Figure 1.4: Time-evolution cue on the left and a loose classroom sketch on the right. Only the Schrödinger-equation fragment is mathematically secure enough to use directly; the loop drawing is kept as blackboard context rather than promoted to a precise formal diagram.

If one changes $t \to -t$ alone, the sign of the left-hand side changes, but the right-hand side does not automatically follow. What repairs the transformation is complex conjugation. Time reversal in quantum mechanics is therefore not just $t \mapsto -t$; it also carries ψ to its complex conjugate, which reverses the sign of the i.

Once this machinery is in place, the physical conclusion is sharp. If particles and antiparticles were exact mirror images in the laws of physics, then baryon-number violation would still not give a directed excess. We need a bias. The lecture states this as failure of C or, more realistically, CP.

Experimentally that bias is real. The lecture uses neutral B-meson decays as the clean illustrative case: a particle and its antiparticle can decay into corresponding channels with measurably different rates. The exact decay products are not the point here. The point is that the world does contain particle-antiparticle asymmetry in its microscopic laws.

1.4.1 Question & Answer

If proton and antiproton total decay rates are constrained to agree, how can an asymmetry still arise?

Because the lecture's answer has two layers.

First, the theorem concerns the *total* decay rate, not the probability of each individual channel. In schematic form,

$$\Gamma_{\text{tot}}(p) = \Gamma_{\text{tot}}(\bar{p}), \tag{1.11}$$

while it still allows different branching among channels, for example

$$\Gamma(p \to e^+ + \gamma) \neq \Gamma(\bar{p} \to e^- + \gamma). \tag{1.12}$$

The missing probability can be compensated in other channels. A proton might also decay to $\mu^+ + \gamma$, while an antiproton decays to $\mu^- + \gamma$, and the total sums can still match exactly.

Second, the lecture is careful to say that the equality theorem is a statement about isolated particles at zero temperature. Baryogenesis does not happen in that environment. It happens in a hot plasma where particles are constantly being kicked around and supplied with extra energy. So even before the deeper equilibrium argument, we should not imagine that the cold-vacuum theorem settles the cosmological problem.

1.5 CPT, Thermal Equilibrium, and the Third Sakharov Condition

Now comes the deepest obstruction. Even if baryon number can change, and even if particle-antiparticle symmetry is violated, a hot universe in thermal equilibrium still cannot maintain a net proton-antiproton excess.

The lecture's reasoning is the following. If the universe were just a hot static pot, then forward time and backward time would be physically equivalent for the microscopic bookkeeping relevant here. In other words, thermal equilibrium carries no arrow of time. But if T is effectively good in that sense, and CPT is an exact theorem, then for present purposes one is driven back to effective CP symmetry. The net result is that equilibrium pushes the system toward equal populations:

$$\text{thermal equilibrium} + CPT \Longrightarrow N_p = N_{\bar{p}}. \tag{1.13}$$

This is the content of the theorem that Susskind emphasizes: in thermal equilibrium one cannot sustain a proton excess over antiprotons.

1.5.1 Question & Answer

Why does thermal equilibrium kill baryogenesis, and what escapes the theorem?

Equilibrium kills baryogenesis because it removes the distinction between forward and backward time. If there is no time-direction in the state of the universe, then the T in CPT does not supply any effective asymmetry, and the particle-antiparticle bias needed for baryogenesis collapses.

What escapes the theorem is the actual early universe. It was not a static hot pot. It was expanding and cooling. Once the expansion is fast enough that microscopic reactions do not fully readjust to equilibrium, forward and backward time become physically distinguishable. That is Sakharov's third condition.

The lecture therefore assembles the three conditions in order:

1. baryon-number violation,
2. violation of particle-antiparticle symmetry, usually phrased as C or CP violation,
3. departure from thermal equilibrium.

Susskind's judgment is that these three ingredients are not only necessary but very likely sufficient in structure. They give baryon number the freedom to change, a bias that pushes the change in one direction, and a cosmological environment in which backward-time equilibrium bookkeeping no longer neutralizes the effect.

1.6 What Baryogenesis Explains, and What It Still Does Not

The lecture now returns to the concrete thermal history. At sufficiently high temperature, photon collisions can create proton-antiproton pairs:

$$\gamma + \gamma \leftrightarrow p + \bar{p}, \qquad E_{\gamma\gamma} \gtrsim 2m_p \approx 2\,\mathrm{GeV}. \tag{1.14}$$

As long as enough high-energy photons are available, the forward and backward reactions occur together in equilibrium. But when the temperature falls below the pair-creation threshold, the left-to-right process becomes inefficient. Proton-antiproton annihilation into photons continues, while recreation from the thermal bath is no longer fast enough to compensate. Almost all the pairs disappear, and only the tiny excess remains.

The lecture only locates this freeze-out era loosely: it is well before nucleosynthesis, at temperatures of order the GeV scale, perhaps seconds into cosmic history. The precise timing is not the point. The point is that once the annihilation becomes one-way in practice, the tiny excess is left behind.

That is why the opening question was upside down. We do not need to explain how a tiny amount of matter manufactured a huge bath of photons. We need to explain why the annihilation did not remove *everything*. Once the excess is established, it is the leftover proton number we observe today.

The lecture also uses this discussion to answer another natural question: if there are so many microwave photons today, why is radiation dynamically negligible now? The answer is the different dilution laws:

$$\rho_{\mathrm{rad}} \propto a^{-4}, \qquad \rho_{\mathrm{m}} \propto a^{-3}. \tag{1.15}$$

Radiation loses one extra power of the scale factor because its wavelengths redshift as the universe expands. Thus the photons from early annihilations are still with us, but their contribution to the present energy budget is small.

There is one more bookkeeping point that the lecture does not want us to miss. If a proton disappears into a positron and a photon, electric charge conservation ties the baryon excess to the lepton excess. That is why one can slide between baryogenesis and leptogenesis in the discussion. The word *baryogenesis* is, as Susskind says, partly a historical relic.

Two further cautions belong here. First, the lecture corrects itself: the standard model does allow baryon violation at some level, though that does not mean it gives a satisfactory quantitative account of the cosmic excess. Second, the oft-repeated claim that standard-model CP violation is too small may well be right, but Susskind's own emphasis is methodological: he is skeptical that the full calculation is under sufficiently good control to be taken as the last word.

What, then, has been explained? The structural answer is clear. We understand the kinds of ingredients the world must contain. What has not been explained is the number itself. The lecture is emphatic about this point: we do not know enough about the very early universe, the relevant high-energy particle content, or the detailed sources of CP violation to calculate the observed 10^{-8} from first principles.

At that point the lecture closes the baryogenesis discussion outright. And then the same logical pattern returns in a different key. Once again a small number forces a vague cosmological question into a precise one.

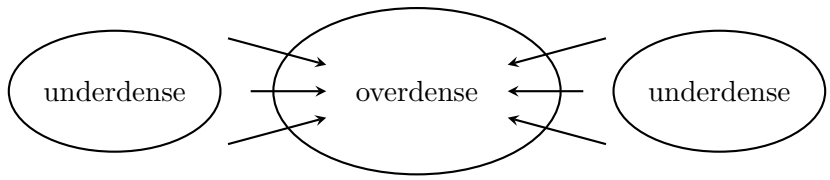

Figure 1.5: A schematic version of the lecture's point about gravitational instability: matter is drawn out of underdense regions and into overdense ones, so small inhomogeneities grow rather than diffuse away.

1.7 A Second Quantitative Puzzle: Homogeneity, Ripples, and Hubble Friction

The next problem is not matter over antimatter but homogeneity. After the cosmic microwave background was measured with high precision, the statement that the universe is nearly isotropic ceased to be an aesthetic slogan and became a quantitative fact. The lecture again begins with a broad question—why is the universe so smooth?—and then tightens it into a numerical one.

The first step is to notice that gravity amplifies lumps rather than erasing them. An overdense region attracts more matter, thereby becoming more overdense still, while its surroundings become more underdense. Gravity does the opposite of diffusion.

Therefore the universe had to be much more homogeneous in the past than it is now. Yet it could not have been *exactly* homogeneous, for then it would have remained exactly homogeneous and no galaxies would ever have formed. The lecture introduces the dimensionless measure of lumpiness

$$\frac{\delta\rho}{\rho}, \tag{1.16}$$

where $\delta\rho$ is the characteristic excess density of a lump above the background. By running the theory backward, early cosmologists estimated that at decoupling this number had to be of order

$$\frac{\delta\rho}{\rho} \sim 10^{-5}. \tag{1.17}$$

Here again the important thing is not that we know why the number is 10^{-5}; the lecture explicitly says we do not. The important thing is that there *is* a number. Once that is known, the homogeneity problem becomes sharp.

The lecture is also careful on a subtle point. It begins to say that homogeneity would imply isotropy and then corrects itself. The safer statement is the one Susskind actually wants: on sufficiently large scales the universe is very smooth and very nearly the same in all directions, but the exact relation between homogeneity and isotropy should not be overstated.

The inflationary answer, in broad outline, is that the universe expanded by an enormous factor and stretched whatever initial roughness it had so thin that on the observable scales it became extraordinarily smooth. Susskind uses the image of blowing up a crinkled balloon: expansion stretches and flattens. He also says something important for the transition into the mathematics: during the inflationary era, temperature is not the main actor. The central fact is the exponential expansion itself.

At this point the lecture makes the bold move that historically sounded crazy: posit one more field in nature, a scalar field ϕ, the inflaton. It is introduced frankly as a field made up in order

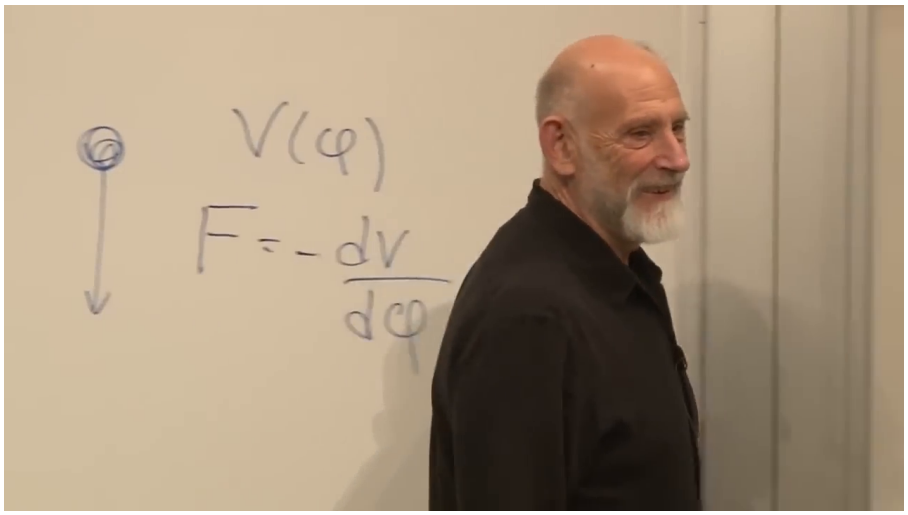

Figure 1.6: Potential and force in the blackboard setup for the falling-stone analogy. This is the visual bridge from ordinary mechanics to the inflaton equation of motion.

to explain the homogeneity and isotropy of the universe. The field is taken to be approximately uniform in space. Then, because the full inflationary equations are deferred to the next lecture, Susskind turns to the one piece of mechanics he wants us to carry away tonight: friction.

The falling stone. Take a stone moving through a viscous fluid, and call its vertical coordinate ϕ. This is a slightly perverse notation for the height of a stone, as Susskind says, but it is the right notation for the field that will soon replace the stone. Suppose the potential energy is $V(\phi)$. Then the force is

$$F = -\frac{dV}{d\phi}. \tag{1.18}$$

If we take the mass to be 1, the frictionless equation is simply

$$\ddot{\phi} = F. \tag{1.19}$$

To model viscosity we add a drag force proportional to velocity and opposite to the motion. With the sign convention that ϕ increases upward, a downward force has $F < 0$, and a stone moving downward has $\dot{\phi} < 0$. The cleaned equation is therefore

$$\ddot{\phi} + \gamma\dot{\phi} = F, \tag{1.20}$$

where $\gamma > 0$ is the drag coefficient. When the acceleration has died away, the motion approaches terminal velocity:

$$\dot{\phi}_{\text{term}} = \frac{F}{\gamma}. \tag{1.21}$$

Because $F < 0$ for downward motion, the terminal velocity is negative, exactly as it should be.

From particle mechanics to a scalar field. The lecture now replaces the stone by a spatially homogeneous scalar field, the inflaton $\phi(t)$. Because the field is taken to be uniform in space, the

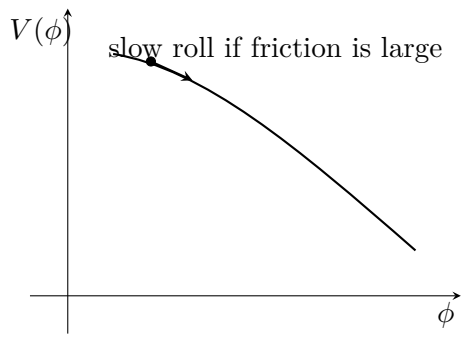

Figure 1.7: A shallow potential hill in the cleaned analogy. Large drag keeps the motion slow even when the force points downhill.

spatial-gradient contribution to the energy is dropped. The homogeneous energy density is then, in the standard cleaned form,

$$\rho_\phi = \frac{1}{2}\dot{\phi}^2 + V(\phi). \tag{1.22}$$

To convert density into total energy, the lecture follows the field inside a comoving unit box. Its physical volume is $a^3(t)$—Susskind writes this as $A^3(t)$ on the board, but we use the standard FRW notation $a^3(t)$. Thus

$$E_\phi = a^3\left(\frac{1}{2}\dot{\phi}^2 + V(\phi)\right). \tag{1.23}$$

The corresponding Lagrangian is

$$L = a^3\left(\frac{1}{2}\dot{\phi}^2 - V(\phi)\right). \tag{1.24}$$

Now apply the Euler–Lagrange equation:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\phi}}\right) = \frac{\partial L}{\partial \phi}, \tag{1.25}$$

$$\frac{\partial L}{\partial \dot{\phi}} = a^3\dot{\phi}, \tag{1.26}$$

$$\frac{\partial L}{\partial \phi} = -a^3\frac{dV}{d\phi}. \tag{1.27}$$

Therefore

$$\frac{d}{dt}\left(a^3\dot{\phi}\right) = -a^3\frac{dV}{d\phi}. \tag{1.28}$$

Expanding the time derivative gives

$$a^3\ddot{\phi} + 3a^2\dot{a}\,\dot{\phi} = -a^3\frac{dV}{d\phi}. \tag{1.29}$$

Divide by a^3, define the Hubble rate by

$$H = \frac{\dot{a}}{a}, \tag{1.30}$$

and we arrive at the lecture's equation of motion:

$$\ddot{\phi} + 3H\dot{\phi} = F, \qquad F = -\frac{dV}{d\phi}. \tag{1.31}$$

Equivalently,

$$\ddot{\phi} + 3H\dot{\phi} + \frac{dV}{d\phi} = 0. \tag{1.32}$$

This is the payoff of the entire friction discussion. The scalar field evolves exactly like the falling stone, except that the viscosity coefficient is now the cosmological quantity $3H$. Hubble expansion acts as a kind of cosmic friction. If H is large and the potential is fairly flat, the field rolls only very slowly. That is the threshold of inflation as the lecture leaves it: not yet the full solution, but the precise equation that will be used next time when the field is coupled back to the Friedmann equation.

1.8 Summary

The lecture is built around two quantitative puzzles. First, the universe contains a baryon excess of only about one part in 10^8, and that forces the Sakharov structure upon us: baryon violation, particle-antiparticle asymmetry, and departure from equilibrium. Second, the universe at decoupling was smooth to about one part in 10^5, and that forces us to ask how such smoothness arose in a world where gravity amplifies lumps.

The baryogenesis part ends deliberately modestly. We know the ingredients that any viable explanation must contain, but we do not know how to compute the observed number from first principles. The inflation part also ends at a threshold rather than a conclusion. We are left with a scalar field in a potential, a Hubble friction term, and the equation

$$\ddot{\phi} + 3H\dot{\phi} + \frac{dV}{d\phi} = 0.$$

That is exactly where the lecture wants to stop: with the sense that the mathematics is now in place, and that next time the universe itself will be allowed to respond to a field rolling very, very slowly downhill.